Roman Korol

Université de Sherbrooke
Department of Chemistry
Office: D1-3033
Phone: 819-821-8000, ext. 61010
Sherbrooke, QC J1K 2R1

Roman.Korol@USherbrooke.ca

Current Position

Université de Sherbrooke, Sherbrooke, Quebec, Canada

Assistant Professor, 2026–present

Education

University of Rochester, Rochester, New York

Postdoctoral Fellow, 2023–2025

California Institute of Technology, Pasadena, California

Ph.D. in Theoretical Chemistry, 2018–2023

University of Toronto, Toronto, Canada
B.Sc. (Honours) in Chemistry with High Distinction, 2014–2018

Fellowships and Awards

Steadman Award	2024
Student Leader Award	2024
Gray-Hill Award Lecture	2023
Patricia Beckman Graduate Fellowship	2018
Michael Rebryk Memorial Scholarship	2018
Ivan Szak Scholarship in Chemistry	2018
St. Michael's College Silver Medal	2018
University of Toronto Excellence Award	2018
St. Michael's College In-Course Scholarship	2018
Canadian Society for Chemistry Silver medal	2018
CQIQC Undergraduate Summer Reaserch Program	2017
University of Toronto Excellence Award	2017
F. E. Beamish Scholarship in Chemistry	2017
Buduchnist Credit Union Scholarship	2017
Ivan Szak Scholarship in Chemistry	2017
Michael Both Award for Outstanding Commitment to Dance	2017
John Melady Memorial Scholarship	2017
C. W. Burton In-Course Scholarship	2017
Gollop Memorial Undergraduate Scholarship in Chemistry	2017
Dean's List Scholar	2017
University of Toronto Excellence Award	2016
Kupcinet-Getz Research Scholarship	2015
University of Toronto Mississauga Honour Roll	2015
Erindale Admission Scholarship	2014
Scholarship of the President of Ukraine *(awarded annually to approximately 250 highest achieving high school students out of over a million)*	2014
Lutsk's student of the year *(awarded annually to the highest achieving high school graduate out of approximately 10,000)*	2014
First Prize at Intel-Eco Ukraine 2014, the national stage of Intel ISEF	2014
Gold medal at the International Ecology Project Olympiad	2013

PUBLICATIONS

14. Tiwari, V; Korol, R; Franco*, I Robust Purely Optical Signatures of Floquet States in Laser-Dressed Crystals. *Phys. Rev. Lett.* **2025**, 135 (18), 186901. DOI: 10.1103/5ywx-7dbs
13. Korol*, R; Chen, X; Franco*, I High-Frequency Tails in Spectral Densities. *J. Phys. Chem. A* **2025**, 129 (15), 3587-3596. DOI: 10.1021/acs.jpca.5c00943
12. Turner*, A C; Korol, R; Bill, M; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 2: Experimental determination of hydrogen isotopic equilibrium for ethane-H_2 from 30 to 200 °C and propane-H_2 from 75 to 200 °C. *Geochim. Cosmochim. Acta* **2025**, 396, 91-106. DOI: 10.1016/j.gca.2025.02.033
11. Korol*, R; Turner, A C; Nandi, A; Bowman, J M; Goddard, W A; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 1: Path-integral calculations with CCSD(T) quality potentials. *Geochim. Cosmochim. Acta* **2025**, 396, 71-90. DOI: 10.1016/j.gca.2025.02.028
10. Turner, A C; Korol, R; Eldridge, D L; Bill, M; Conrad, M E; Miller, T F; Stolper*, D A Experimental and theoretical determinations of hydrogen isotopic equilibrium in the system CH_4-H_2-H_2O from 3 to 200 °C. *Geochim. Cosmochim. Acta* **2021**, 314, 223-269. DOI: 10.1016/j.gca.2021.04.026
9. Korol, R; Rosa-Raíces, J L; Bou-Rabee, N; Miller*, T F Dimension-free path-integral molecular dynamics without preconditioning. *J. Chem. Phys.* **2020**, 152 (10), 104102. DOI: 10.1063/1.5134810
8. Eldridge, D L; Korol, R; Lloyd, M K; Turner, A C; Webb, M A; Miller, T F; Stolper*, D A Comparison of Experimental vs Theoretical Abundances of $^{13}CH_3D$ and $^{12}CH_2D_2$ for Isotopically Equilibrated Systems from 1 to 500 °C. *ACS Earth Space Chem.* **2019**, 3 (12), 2747-2764. DOI: 10.1021/acsearthspacechem.9b00244
7. Korol, R; Bou-Rabee, N; Miller*, T F Cayley modification for strongly stable path-integral and ring-polymer molecular dynamics. *J. Chem. Phys.* **2019**, 151 (12), 124103. DOI: 10.1063/1.5120282
6. Korol, R; Segal*, D Machine Learning Prediction of DNA Charge Transport. *J. Phys. Chem. B* **2019**, 123 (13), 2801-2811. DOI: 10.1021/acs.jpcb.8b12557
5. Korol, R; Segal*, D From Exhaustive Simulations to Key Principles in DNA Nanoelectronics. *J. Phys. Chem. C* **2018**, 122 (8), 4206-4216. DOI: 10.1021/acs.jpcc.7b12744
4. Korol, R; Kilgour, M; Segal*, D ProbeZT: Simulation of transport coefficients of molecular electronic junctions under environmental effects using Büttiker's probes. *Comput. Phys. Commun.* **2018**, 224, 396-404. DOI: 10.1016/j.cpc.2017.10.005
3. Korol, R; Kilgour, M; Segal*, D Thermopower of molecular junctions: Tunneling to hopping crossover in DNA. *J. Chem. Phys.* **2016**, 145 (22), 224702. DOI: 10.1063/1.4971167
2. Longobardi, L E; Zatsepin, P; Korol, R; Liu, L; Grimme, S; Stephan*, D W Reactions of Boron-Derived Radicals with Nucleophiles. *J. Am. Chem. Soc.* **2016**, 139 (1), 426-435. DOI: 10.1021/jacs.6b11190
1. Korol*, R V; Yanchuk, O M; Marchuk, O V; Orlov, V F; Moroz, I A; Vyshnevskyi, O A Size Stabilizers in Two-electrode Synthesis of ZnO Nanorods. *Phys. Chem. Solid St.* **2021**, 22 (2), 380-387. DOI: 10.15330/pcss.22.2.380-387

Presentations and Awards

CERMM Seminar, Concordia University, Montreal, Quebec — 2026
Invited talk: "Quantum Effects in Complex Systems"

"Lunch and Learn" series, Department of Chemistry, Université de Sherbrooke, Quebec — 2026
Talk: "Quantum Effects in Complex Systems"

CCQS Seminar at the Physics department, University of Rochester, Rochester, New York — 2025
Talk: "How NOT to build control-target gates in semiconductor quantum dots and beyond"

Summer of Molecular Quantum Information Science Seminars Series — 2025
Talk: "Opening quantum dynamics on gate-defined quantum dots with RLC circuits"

Rochester Conference on Coherence and Quantum Science, Rochester, New York — 2025
Poster: "Hybrid singlet-triplet qubits enable fault-tolerant single-pulse CNOT gate"

American Conference on Theoretical Chemistry, Chapel Hill, North Carolina — 2024
Poster: "Analog Simulation of Open Quantum Dynamics"

Gray-Hill lecture at the Occidental college, Los Angeles, California. — 2023
Award talk: "A window to Earth's past with the help of theoretical chemistry"

Canadian Chemistry Conference and Exhibition, Calgary, Canada — 2022
Contributed talk: "Accurate quantum statistics from improved path-integrals in imaginary time"

Molecular Science Mini-meeting, Montreal, Canada — 2022
Poster: "Dimension-free ring-polymer molecular dynamics"

ACS Spring meeting, San Diego, California — 2022
Poster: "Accurate quantum statistics from improved path-integrals in imaginary time"

Geological and Planetary Sciences seminar at Caltech, Pasadena, California — 2022
Invited talk: "D and ^{13}C exchange equilibria using Path-Integral Monte-Carlo"

Berkeley Statistical Mechanics Meeting — 2020
Poster: "Cayley modification for strongly stable path-integral molecular dynamics"

CECAM BioMolecular Electronics Conference, Madrid, Spain — 2018
Poster: "Principles of Charge Transport in DNA: from extensive simulations to neural networkds"

28$^{\text{th}}$ Canadian Symposium on Theoretical and Computational Chemistry, Windsor, Canada — 2018
Poster prize: "Charge transport in DNA: From comprehensive simulations to key principles"

100$^{\text{th}}$ Canadian Chemistry Conference, Toronto, Canada — 2017
Poster prize: "Tunneling to Hopping Crossover in Thermopower of DNA Molecular Junctions"

Chemical Biophysics Symposium, Toronto, Canada — 2017
Contributed talk: "DNA Molecular Junctions: Tunneling to Hopping Crossover"

33$^{\text{rd}}$ Symposium on Chemical Physics, Waterloo, Canada — 2017
Contributed talk: "Probing mechanisms of charge transport in DNA with Landauer-Büttiker formalism"

45th Southern Ontario Undergraduate Student Chemistry Conference, Toronto, Canada — 2017
1$^{\text{st}}$ prize talk: "Tunneling to Hopping Crossover in DNA & DNA-like molecular junctions"

Service

	Departmental Graduate Studies Committee (DGSC)	2026–present
	Student monitoring committees (3 students)	2026–present
	Student Representative at the Chemistry Department Advisory Committee	2016–2017
	2nd year representative at the Chemistry student union	2016–2017
	Board member of the *Chemistry Connections* student group	2015–2016

Membership

Canadian Society for Chemistry (CSC)
Physical, Theoretical and Computational Chemistry (PTC) Division of the CSC
Canadian Association of Theoretical Chemists (CATC)
Centre for Research in Molecular Modeling (CERMM)
Centre for the Study of Advanced Materials for Batteries and Renewable Energy Storage Systems of the Université de Sherbrooke (CÉMAPROVUS)

Community Volunteer Initiatives

Spearheaded and coordinated humanitarian supplies shipment to Ukraine — 2022–2023
  from Caltech campus and beyond

Volunteer at the Nova Ukraine non-profit, Stanford, California — 2021–2023
  Website development, established and coordinated partnership with *Teach for Ukraine*

Volunteer at the Teach for Ukraine nonprofit, Kyiv, Ukraine — 2021–2022
  Recruited and interviewed candidate teachers at the remote interview stage

International student orientation leader — 2019, 2020

"Big sibling" mentor for the incoming graduate students at Caltech — 2019, 2020

Science outreach program volunteer through Caltech Y — 2018–2020

High-school tutoring with CAUSE Tutoring non-profit — 2018–2019

University of Toronto Peer Tutoring group tutor — 2015–2018

Teaching

Course development: Computational chemistry labs, Chem3 at Caltech — 2022
Focus on structure-function relations and the dangers of approximations.

High School Teacher, Rotman Arts and Science School, Vaughan, Canada — 2020–2022
  Academic stream, grade 11 and 12 Chemistry, Grade 10 Science.
  Student placed 3rd in Vaughan, top 200 in Canada at the Avogadro chemistry contest

International Chemistry Olympiad Coach
  Canadian National Team (4 students) – 2 bronze, 1 silver medals — Spring 2018
  Ukrainian National team (1 student) – bronze medal — Spring 2014

Private tutoring of Chemistry, Physics and Math — 2014–2018
  High school students: accepted to University college (UK), Columbia University (USA) and others

Chemical Biology summer school, Lutsk, Ukraine — Summer 2015
  Designed problems and experiments to help high school students master key concepts in chemistry

Peer Review

ACS journals: ACS Physical Chemistry Au
APS journals: Physical Review Letters, Physical Review A, B, E and Research
Elsevier journals: Organic Geochemistry, Chemical Geology
RSC journals: Physical chemistry chemical physics
Wiley journals: Rapid communications in mass spectrometry

SUMMER SCHOOLS AND WORKSHOPS	Condensed Phase Dynamics Workshop at TSRC (Virtual)	2020
	Theoretical Chemistry School at TSRC, Telluride, Colorado	2019
	Weizmann Institute of Science, Rehovot, Israel Kupcinet-Getz Scholar at Rubtchinski lab	2015
PAST EMPLOYMENT	High School Teacher, Rotman Arts and Science School, Vaughan, Canada	2020–2022
	Research Assistant, Department of Linguistics, University of Toronto Heritage language variation and change project	2016–2018
EXTRA CURRICULARS	Rock clibmbing, weightlifting	since 2014
	Guitar, base	since 2012
	Ukrainian folk dance	since 2004
LANGUAGES	Fluent in Ukrainian, English & Russian	
COMPUTER LANGUAGES	Python, C++, MATLAB, FORTRAN, Mathematica, Bash; Web development (PHP & django, HTML, CSS, JS)	